



AI LMS (INTELLIGENCE LEARNING NETWORK)

The domain of the Project:

Artificial Intelligence / EdTech - AI-Powered Learning Management System

AI LMS PROJECT

Team Mentor (and designation):

Mr. Shrikant Subarmaniyam

Team Members:

Mr. Shiva Singh

Period of the project

March 2026 to September 2026



Declaration

The project titled “AI LMS (Intelligence Learning Network)” has been mentored by Mr. Shrikant Subarmaniyam, organised by SURE Trust, from March 2026 to September 2026, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below have worked on it successfully and enhanced their practical knowledge in the domain.

Team Members:

Mr. Shiva Singh — Signature

Mentor's Name
Designation — Company Name

Shrikant Subarmaniyam
Mentor, SURE Trust

Prof. Radhakumari
Executive Director & Founder
SURE Trust



Table of Contents

1. Executive Summary
2. Introduction
3. Project Objectives
4. Methodology & Results
5. Social / Industry Relevance of the Project
6. Learning & Reflection
7. Future Scope & Conclusion



1. Executive Summary

The AI LMS (Intelligence Learning Network) is a next-generation, AI-driven educational platform engineered to transform traditional learning into an interactive, multimodal experience. The system synthesizes grade-tailored lesson plans, generates multilingual audio voiceovers and video storyboards, converts OCR handwritten notes into digital text, creates interactive quizzes with automatic grading, tracks student telemetry in a cloud PostgreSQL database, and builds personalized study roadmaps.

By combining nine specialised AI intelligence modules within a single ecosystem, the project delivers a comprehensive, end-to-end solution that supports students and educators across content creation, assessment, doubt resolution, and progress tracking — all through one intuitive interface. The platform demonstrates how generative AI, large language models, text-to-speech, computer vision (OCR), and cloud database technologies can be integrated into a practical, deployable EdTech product.

Key highlights for stakeholders include automated multilingual content generation (11+ languages including Hindi, Punjabi, and Hinglish), real-time analytics dashboards for student performance, and automated PDF report generation — collectively reducing manual effort for teachers while improving personalised learning outcomes for students.

2. Introduction

Background and Context

Traditional Learning Management Systems are largely static repositories of content, offering limited personalisation and requiring significant manual effort from educators to create lesson material, assessments, and progress reports. With the rapid advancement of generative AI, large language models, and multimodal content generation APIs, there is a significant opportunity to build a learning platform that actively creates, adapts, and evaluates educational content rather than simply hosting it.

Problem Statement and Goals

The project set out to build a single ecosystem that could: generate structured lesson content on demand; convert handwritten or scanned material into usable digital text; produce multilingual audio and video learning material; assess students automatically through quizzes and subjective evaluation; and give both students and educators clear, data-backed visibility into learning progress.

Scope and Limitations

The scope covers nine core modules spanning lesson generation, multimedia studio production, OCR-based homework conversion, quiz creation and grading, a doubt-resolution chatbot, assignment evaluation, roadmap generation, analytics, and study planning. The current implementation relies on third-party AI APIs (Gemini, Imagen 3, ElevenLabs, Hugging Face), which introduces dependency on external service availability, rate limits, and free-tier constraints such as inference queue delays.

Innovation Component

The project's innovation lies in unifying multiple distinct AI capabilities — text generation, image generation, video synthesis, multilingual text-to-speech, OCR, and automated grading — into one cohesive student-facing platform, backed by a cloud-native PostgreSQL database for persistent telemetry and analytics.



3. Project Objectives

Defined Objectives

- Build an AI Lesson Generator that produces structured, grade-tailored lesson notes, diagrams, and printable PDFs.
- Develop a Multilingual Audio & Video Studio capable of producing narrated audio, short video clips, and styled PowerPoint decks in 11+ languages.
- Implement an AI Homework & OCR Converter to digitise handwritten notes and scanned textbook pages.
- Create an Interactive Quiz Creator with mandatory student identification and automatic score computation.
- Provide a 24/7 Student Doubt Chatbot powered by an LLM for conversational academic support.
- Build an Assignment Evaluator that grades subjective answers against model rubrics with constructive feedback.
- Generate personalised 14-day Learning Path Roadmaps targeting each student's weak subjects.
- Deliver Progress Telemetry Analytics with searchable, exportable PDF performance reports.
- Implement an AI Study Planner producing daily revision timetables based on available study hours and exam dates.

Expected Outcomes and Deliverables

- A fully functional full-stack web application (React frontend + FastAPI backend).
- A cloud-hosted PostgreSQL database recording quiz results and telemetry.
- Nine integrated AI modules accessible through a unified dashboard.
- Automated PDF and PPTX export capability for lessons, progress reports, and presentations.

4. Methodology & Results

Methods / Technology Used

The system follows a modular client-server architecture. A React.js single-page application communicates with a FastAPI backend over REST endpoints; the backend orchestrates calls to multiple generative-AI providers and persists structured data to a cloud PostgreSQL database.

Layer	Technology / Tools Used
Frontend	React.js with Vite, react-router-dom; custom GuardianEye Cyber-HUD design system (HSL neon glows, dark theme); Recharts for analytics; jsPDF / html2canvas for client-side PDF export
Backend	FastAPI (Python 3.14) with async endpoints and CORS middleware; SQLAlchemy ORM connected to a Neon Serverless PostgreSQL database
Generative AI	Google Gemini API (gemini-2.5-flash) via LangChain / Google GenAI SDK; Google Imagen 3 for image generation
Video & Presentation	Hugging Face Inference API (damo-vilab/text-to-video-ms-1.7m), moviepy, Pillow, python-pptx



Text-to-Speech	ElevenLabs Multilingual v2 API (eleven_multilingual_v2) with gTTS fallback
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Project Architecture

User requests originate from the React frontend and are routed to the FastAPI backend engine, which dispatches them to one of nine AI intelligence modules: (1) AI Lesson Generator, (2) Audio & Video Studio, (3) AI Homework & OCR Converter, (4) Interactive Quiz Creator, (5) Student Doubt Chatbot, (6) Assignment Evaluator, (7) Learning Path Roadmap, (8) Progress Telemetry Analytics, and (9) AI Study Planner. The Quiz Creator and Analytics modules both read from and write to the Neon PostgreSQL database, enabling persistent, queryable student telemetry.

Module Breakdown

- AI Lesson Generator (routers/lesson.py): accepts grade level, subject, and topic to generate structured lesson notes, AI visual diagrams, teacher activity guides, and printable PDF notes.
- Multilingual Audio & Video Studio: synthesizes NotebookLM-style audio overviews (.mp3), Hugging Face text-to-video clips (.mp4), and 16:9 Gamma-style PowerPoint decks (.pptx) in 11+ languages including English, Hinglish, Hindi, and Punjabi.
- AI Homework & OCR Converter (routers/converter.py): extracts text from uploaded PDF textbooks, PNG/JPEG notebook photos, .docx, and .txt files using PyMuPDF, pypdf, and Tesseract OCR.
- Interactive Quiz Creator (routers/quiz.py): enforces mandatory student identification (name, roll number, student ID) and generates MCQ, true/false, and fill-in-the-blank questions with automated score computation.
- Student Doubt Chatbot (routers/chatbot.py): provides 24/7 conversational academic support using the Gemini LLM.
- Assignment Evaluator (routers/evaluate.py): automatically grades subjective answers against model rubrics, returning numeric scores out of 10 with constructive feedback.
- Learning Path Roadmap (routers/learning_path.py): generates personalised 14-day adaptive study roadmaps targeting weak subjects.
- Progress Telemetry Analytics (routers/analytics.py): queries student records by name, roll number, or ID; computes average scores, pass/fail counts, and exports formal PDF progress reports.
- AI Study Planner (routers/study_planner.py): generates daily revision timetables customised to available study hours and target exam dates.

Key Challenges Encountered & Engineering Solutions

Challenge	Root Cause	Engineering Solution
PyMuPDF import error (No module named fitz)	Missing native binary library in the virtual environment during PDF processing	Installed PyMuPDF and pypdf into the backend venv; built dual fallback PDF parsing logic in routers/converter.py
Database column mismatch on score save	Initial QuizResult table lacked student_name and roll_no columns	Updated SQLAlchemy schema, dropped legacy tables, and safely migrated the new schema to Neon PostgreSQL
Hugging Face text-to-video timeouts (503)	Free-tier inference models experience queue delays during cold starts	Built a multi-model fallback loop (damo-vilab, ali-vilab, ZeroScope) with an



		automated moviepy slide-animation fallback
Regional script font rendering (Hindi / Punjabi)	Pillow's default bitmap fonts cannot render Indic/Unicode text on video frames	Implemented dynamic system font loading (Nirmala.ttf, Raavi.ttf, Mangal.ttf, Segoe UI) for multi-language rendering
Audio / video track synchronisation	Audio duration mismatch when attaching ElevenLabs voiceover to fixed video clips	Used moviepy's AudioClip.duration matching to dynamically sync clip duration to audio duration
Unformatted presentation exports	Default python-pptx bullet layouts looked generic and plain	Re-architected the pptx generator to output 16:9 widescreen, Gamma.app-styled dark cards with embedded visual graphics

Final Project Working Screenshots

The following screenshots illustrate the deployed frontend application (React + Vite, running on localhost:3000) and the backend development environment (FastAPI, Python) used to build the AI LMS platform.

Frontend Screenshots

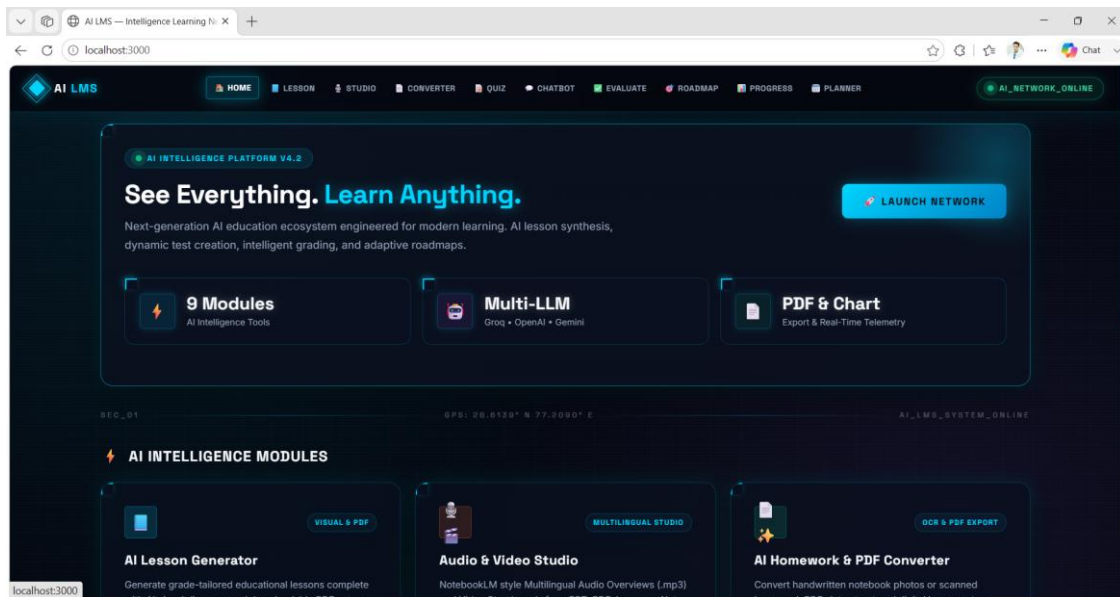


Fig 1: Home page — platform overview showing the 9 AI Intelligence Modules, system status, and quick highlights (9 Modules, Multi-LLM, PDF & Chart export).



Fig 2: AI Lesson Generator — topic input, grade-level selector, and NotebookLM-style audio/video script generation options.

Fig 3: Multilingual Audio & Video Studio — source upload, target class, narration language, and output media mode selection.

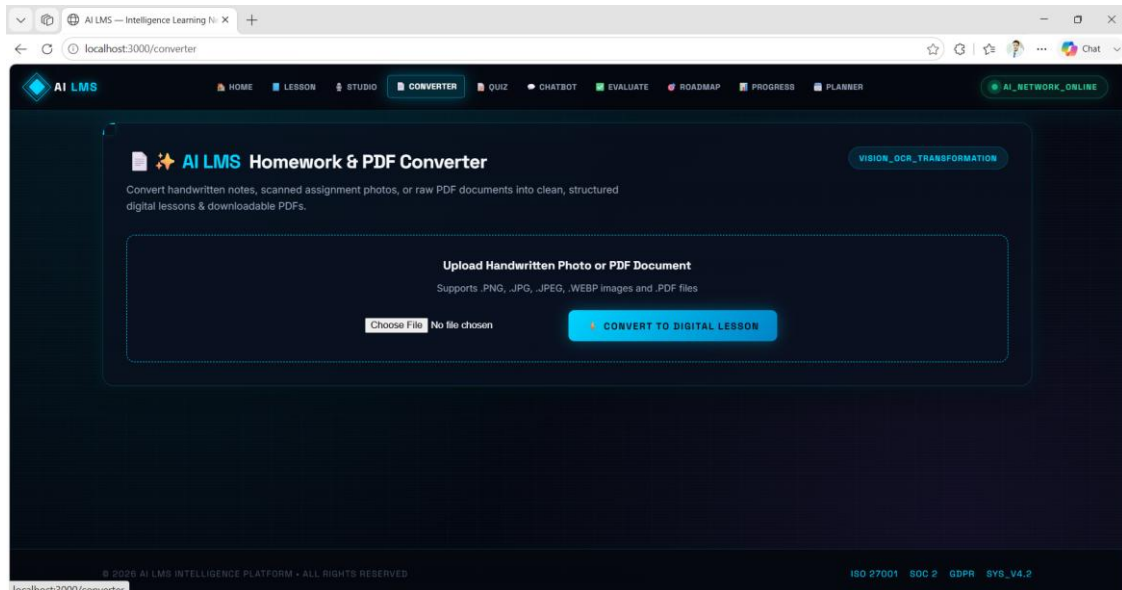


Fig 4: AI Homework & PDF Converter — OCR-based upload interface for handwritten notes, scanned PDFs, and images.

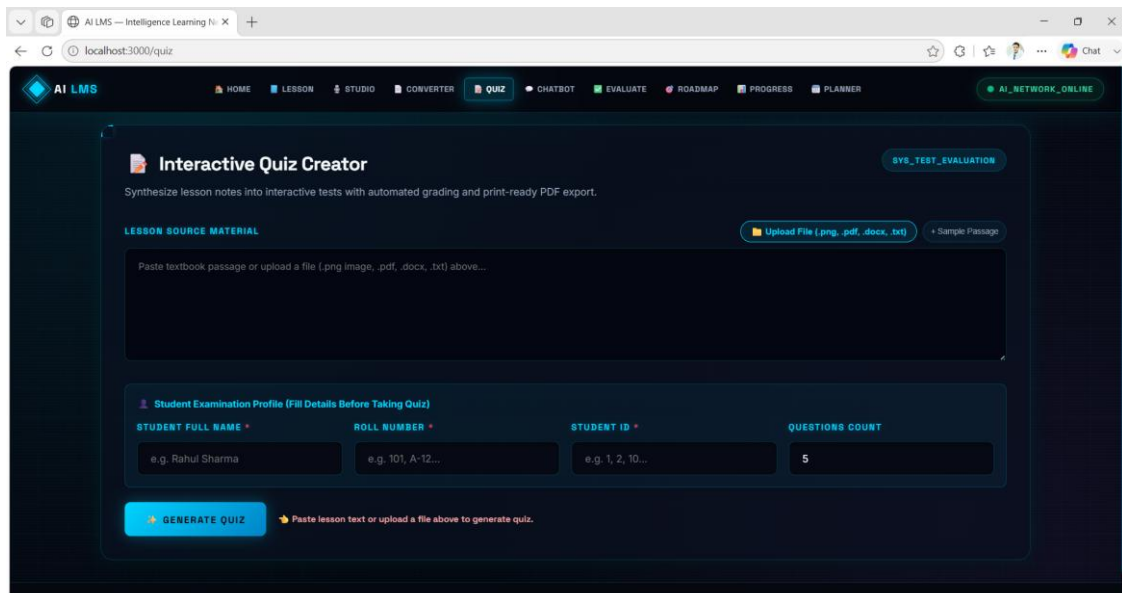


Fig 5: Interactive Quiz Creator — lesson source input, mandatory student examination profile fields, and question-count configuration.

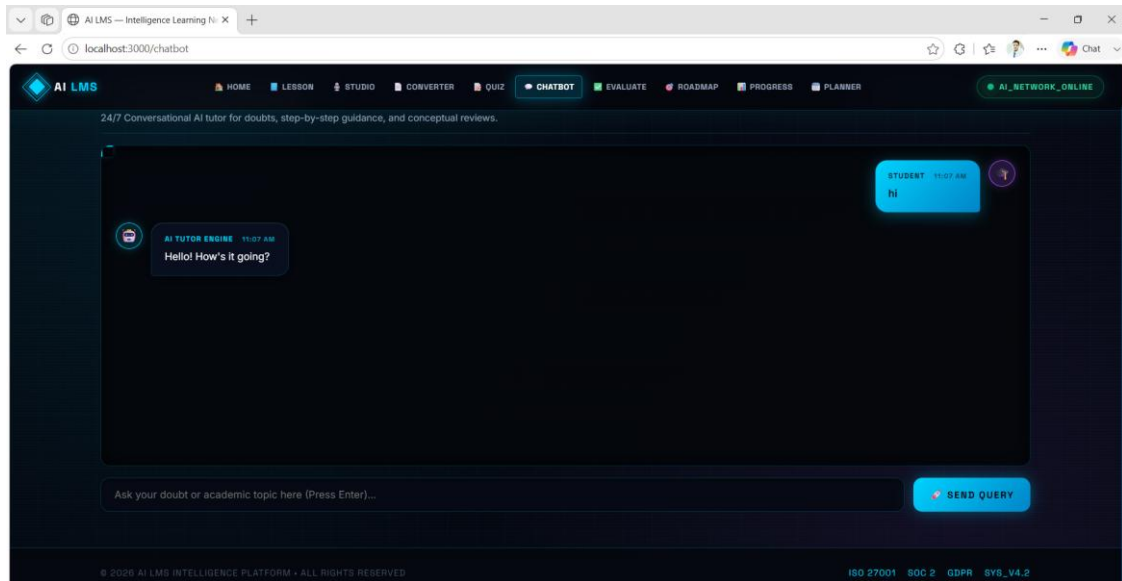


Fig 6: Student Doubt Chatbot — real-time conversational interface powered by the Gemini LLM.

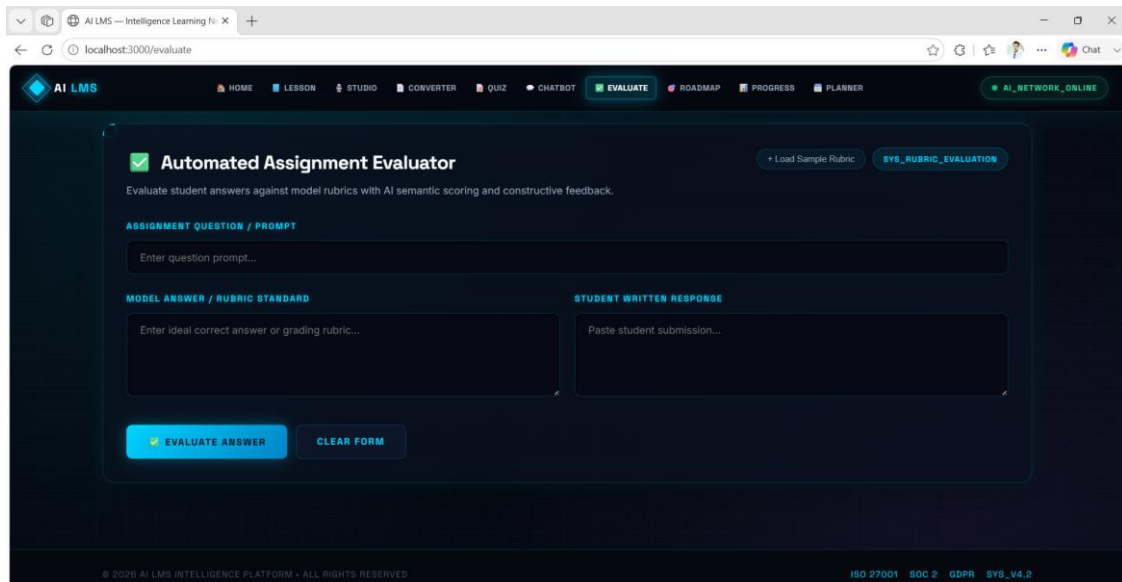


Fig 7: Automated Assignment Evaluator — rubric-based scoring interface comparing model answers with student responses.

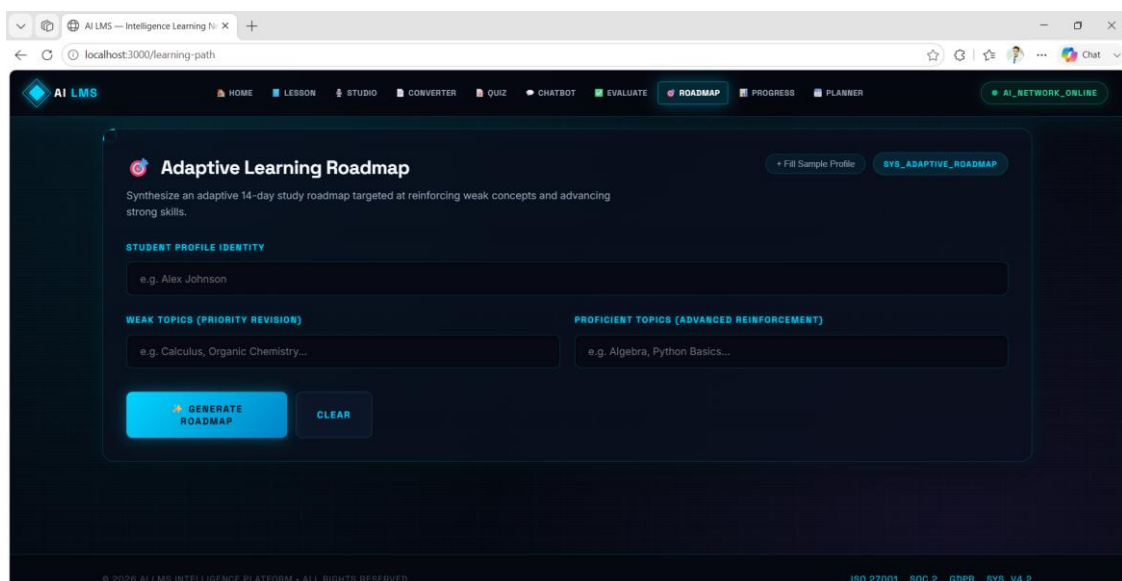




Fig 8: Adaptive Learning Roadmap — 14-day personalised study plan generator based on weak and proficient topics.

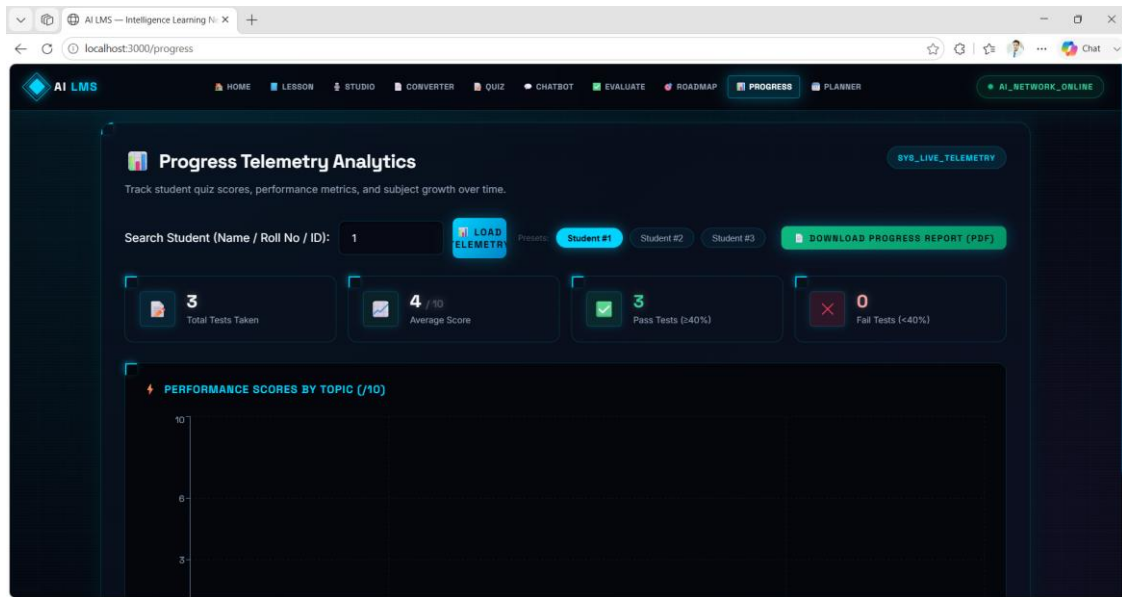


Fig 9: Progress Telemetry Analytics — student search, performance metrics (tests taken, average score, pass/fail count), and topic-wise score charts.

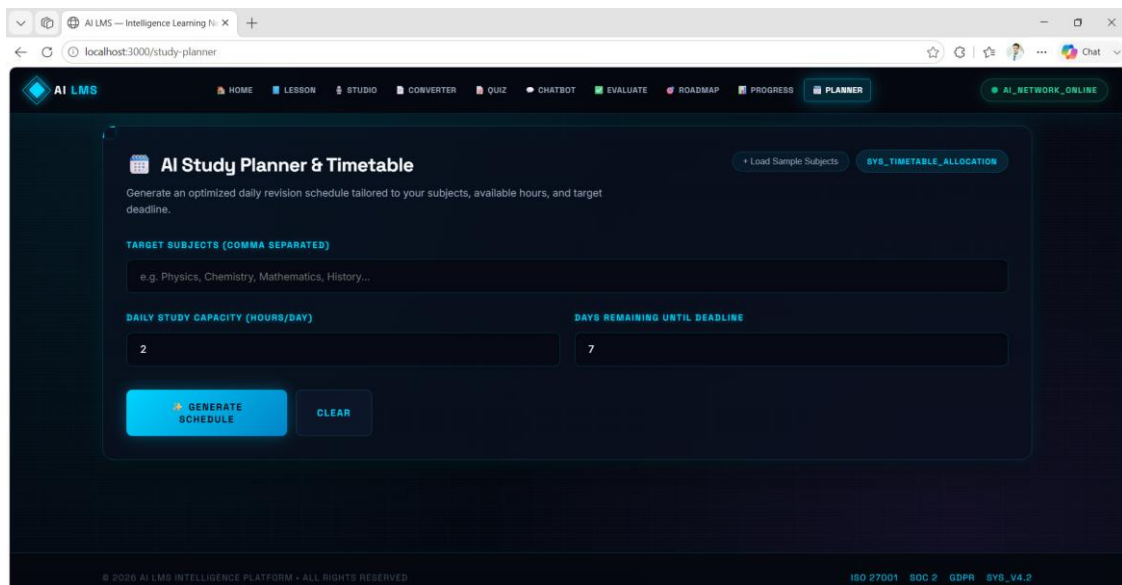


Fig 10: AI Study Planner — daily revision timetable generator based on target subjects, study hours, and deadline.

Backend Development Screenshot



```
1 from fastapi import FastAPI, Request, HTTPException
2 from fastapi.responses import JSONResponse, StreamingResponse
3 from fastapi.staticfiles import StaticFiles
4 from fastapi.templating import Jinja2Templates
5
6 import os
7 import re
8 import html
9
10 from reportlab.lib.pagesizes import A4
11 from reportlab.lib.styles import getSampleStyleSheet, ParagraphStyle
12 from reportlab.lib.enums import TA_CENTER
13 from reportlab.platypus import (
14     SimpleDocTemplate,
15     Paragraph,
16     Spacer,
17     Image as ReportLabImage,
18 )
19
20 from reportlab.lib.units import inch
21
22 from llm import get_llm
23 from database import get_db, lesson
24 from image_generator import generate_image
25
26 # =====
27 # ROUTER
28
```

Terminal Output:

```
INFO: 127.0.0.1:51738 - "GET /analytics/progress/1 HTTP/1.1" 200 OK
INFO: 127.0.0.1:55458 - "OPTIONS /chatbot/ask HTTP/1.1" 200 OK
D:\project\ai-lms-project\ai-lms-project\backend\venv\Lib\site-packages\langchain_google_genai\chat_models.py:3237: UserWarning: Model 'gemini-1.5-flash-lite' uses fixed sampling defaults; the sampling parameter(s) temperature will be ignored.
  request = self._build_request_config(
INFO: 127.0.0.1:55458 - "POST /chatbot/ask HTTP/1.1" 200 OK
INFO: 127.0.0.1:63185 - "GET /analytics/progress/1 HTTP/1.1" 200 OK
INFO: 127.0.0.1:65882 - "GET /analytics/progress/1 HTTP/1.1" 200 OK
INFO: 127.0.0.1:65882 - "GET /analytics/progress/1 HTTP/1.1" 200 OK
```

Fig 11: Backend codebase (lesson.py) in the IDE, showing FastAPI router imports, ReportLab PDF generation, and live Uvicorn server logs confirming successful API calls (chatbot, analytics).

Project GitHub Link

<https://github.com/Shiva272001>

5. Social / Industry Relevance of the Project

The AI LMS directly addresses a widely recognised gap in the Indian education sector: unequal access to personalised, high-quality learning support, particularly for students in regional-language and rural settings. By generating lesson content, audio, and video in Hindi, Punjabi, and Hinglish alongside English, the platform lowers the language barrier that often limits access to quality digital learning resources.

From an industry standpoint, the project mirrors real-world EdTech product architecture — combining LLM-based content generation, OCR digitisation, automated assessment, and cloud analytics — giving the team direct, hands-on exposure to patterns used by commercial platforms such as adaptive learning systems and AI tutoring products. This aligns with SURE Trust's mission of equipping educated but unemployed rural youth with industry-relevant, hands-on project experience that improves employability.

The automatic grading and analytics modules also reduce the administrative burden on teachers, freeing time for more direct student engagement, while the Assignment Evaluator and Doubt Chatbot extend academic support beyond classroom hours.

6. Learning & Reflection

New Learnings (Technology & Management)

- Hands-on experience integrating multiple generative AI APIs (Gemini, Imagen 3, ElevenLabs, Hugging Face) into a single production-style backend.
- Practical understanding of asynchronous API design using FastAPI, and ORM-based schema migration with SQLAlchemy on a serverless PostgreSQL database (Neon).
- Experience debugging real-world integration issues — missing native libraries, third-party inference timeouts, font-rendering limitations for Indic scripts, and audio/video synchronisation.



- Exposure to automated document generation pipelines using python-pptx, moviepy, and jsPDF/html2canvas for professional, presentation-ready outputs.
- Project and time management skills gained from coordinating a multi-module system with interdependent components (e.g., Quiz Creator and Analytics sharing the same database).

Overall Experience

Building the AI LMS was an end-to-end exercise in translating a broad idea — an AI-powered learning ecosystem — into a working, modular product. The most valuable part of the experience was learning to design fallback strategies (multi-model video generation, TTS fallback, dual PDF parsing) so the system remains reliable even when individual third-party services are slow or unavailable. This project strengthened both technical skills in full-stack and AI engineering, and practical problem-solving skills in debugging production-style issues under real constraints.

7. Future Scope & Conclusion

Recap of Objectives and Achievements

The project successfully delivered all nine planned AI modules — lesson generation, multilingual audio/video studio, OCR-based homework conversion, quiz creation with automated grading, a doubt-resolution chatbot, subjective assignment evaluation, personalised learning roadmaps, progress analytics, and an AI study planner — integrated into a single cohesive platform backed by a cloud PostgreSQL database.

Future Scope

- Phase 1 (Q3 2026): Real-time, low-latency voice conversations using the Gemini Live API, enabling students to speak directly with their AI tutor.
- Phase 2 (Q4 2026): A School & Teacher Analytics Portal with batch quiz assignment and class-wide performance leaderboards.
- Phase 3 (Q1 2027): An offline-capable native mobile app (React Native) with local audio-lesson caching and quiz downloads.
- Phase 4 (Q2 2027): VR/AR interactive science labs featuring 3D molecular and physics manipulatives.
- Additional planned enhancements: a teacher & school management dashboard with gradebook sync and bulk assignment dispatch; personalised AI voice cloning for automated lesson narration; and gamified student badges, XP points, and streaks to boost engagement.

Conclusion

The AI LMS Ecosystem successfully bridges generative AI technology with practical educational needs. By integrating multimodal content generation — text, audio, video, presentations, OCR, and analytics — into one intuitive interface, it provides students and educators with an all-in-one platform for modern, personalised learning, while giving the project team substantial hands-on experience in applied AI engineering.